

**GAGE REPEATABILITY AND REPRODUCIBILITY DATA SHEET
VARIABLE DATA RESULTS**

Part Number 5762	Gage Name Calipers	Appraiser A Bob	Part Number 5762	Gage Name Calipers	Appraiser A Bob
Part Name Door Panel	Gage Number 1019	Appraiser B Cindy	Part Name Door Panel	Gage Number 1019	Appraiser B Cindy
Characteristic Thickness	Specification 5 8	Gage Type 6" Digital Calipers	Characteristic Thickness	Gage Type 6" Digital Calipers	Appraiser C Tom
Characteristic Classification 6.5+/-1.5mm	Trials 3	Parts 10	Appraisers 3	Date Performed	

**GAGE REPEATABILITY AND REPRODUCIBILITY DATA SHEET
VARIABLE DATA RESULTS**

APPRaiser/ TRIAL #	PART										AVERAGE	Measurement Unit Analysis				% Tolerance (Tol)						
	1	2	3	4	5	6	7	8	9	10		Repeatability - Equipment Variation (EV)		Trials	K1	% EV	=					
1. A	1	6.55	6.58	6.63	6.51	6.53	6.58	6.53	6.54	6.53	6.49	6.547	EV	=	R x K ₁	2	0.8865	100 (EV/Tol)	=			
2.	2	6.54	6.60	6.61	6.52	6.52	6.59	6.53	6.54	6.53	6.49	6.547	=	0.010 x 0.5907	3	0.5907	100(0.006/0.500)	=				
3.	3	6.55	6.59	6.62	6.51	6.53	6.59	6.53	6.53	6.54	6.48	6.547	=	0.006			1.22		=			
4.	AVE	6.55	6.59	6.62	6.51	6.53	6.59	6.53	6.54	6.53	6.49	x _a =	6.547	Reproducibility - Appraiser Variation (AV)				% AV	=			
5.	R	0.01	0.02	0.02	0.01	0.01	0.01	0.00	0.01	0.01	0.01	r _a =	0.011	AV	=	{(x _{DIFF} x K ₂) ² - (EV ² /nr)} ^{1/2}		100 (AV/Tol)	=			
6.	B	1	6.55	6.59	6.62	6.51	6.52	6.58	6.52	6.53	6.54	6.48	6.544	=	{(0.001 x 0.5236) ² - (0.006 ² /(10 x 3))} ^{1/2}		100(0.000/0.500)	=				
7.	2	6.55	6.60	6.62	6.52	6.52	6.59	6.54	6.53	6.53	6.49	6.549	=	0.000			0.00		=			
8.	3	6.55	6.60	6.61	6.52	6.52	6.59	6.54	6.54	6.53	6.50	6.550			Appraiser	2	3					
9.	AVE	6.55	6.60	6.62	6.52	6.52	6.59	6.53	6.53	6.53	6.49	x _b =	6.548	n = parts	r = trials	K ₂	0.7087	0.5236	% GRR	=		
10.	R	0.00	0.01	0.01	0.01	0.00	0.01	0.02	0.01	0.01	0.02	r _b =	0.010	Repeatability & Reproducibility (GRR)				100 (GRR/Tol)				
11.	C	1	6.55	6.59	6.62	6.50	6.50	6.58	6.53	6.54	6.53	6.50	6.544	GRR	=	{(EV ² + AV ²) ^{1/2}	Parts	K ₃	=			
12.	2	6.56	6.60	6.61	6.51	6.50	6.59	6.54	6.54	6.53	6.51	6.549	=	{(0.006 ² + 0.000 ²) ^{1/2}	2	0.7087	=	100(0.006/0.500)				
13.	3	6.55	6.59	6.61	6.51	6.51	6.59	6.53	6.53	6.54	6.50	6.546	=	0.006	3	0.5236	=	1.22	Gage system O.K			
14.	AVE	6.55	6.59	6.61	6.51	6.50	6.59	6.53	6.54	6.53	6.50	x _c =	6.546	Part Variation (PV)				4	0.4464	% PV	=	
15.	R	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	r _c =	0.010	PV	=	R _p x K ₃	5	0.4032	100 (PV/Tol)	=		
16.	PART		19.77	19.86	19.54	19.58	19.76	19.59	19.61	19.60	19.46	X=	6.547	=	0.123 x 0.3145	6	0.3745	100(0.039/0.500)				
	AVERAGE	6.55	6.59	6.62	6.51	6.52	6.59	6.53	6.54	6.53	6.49	R _p =	0.123	=	0.039	7	0.3534	7.76				
17.	(r _a + r _b + r _c) / (# OF APPRAISERS) =											R=	0.010	Tolerance (Tol)				8	0.3378	ndc	=	
18.	x _{DIFF} = (Max x - Min x) =											x _{DIFF} =	0.001	Tol	=	Upper - Lower / 6	9	0.3247	=	1.41(PV/GRR)		
19.	* UCL _R = R x D ₄ =											UCL _R =	0.027	=	(8 - 5) / 6	10	0.3145	=	1.41(0.039/0.006)			
													=	0.500					= 8			
* D ₄ =3.27 for 2 trials and 2.58 for 3 trials. UCL _R represents the limit of individual R's. Circle those that are														Gage discrimination acceptable								